

## AERONEXUS TECHNOLOGIES

# Corporate VPN Access and Remote Connectivity Guide

Global Technical Standards, Security Requirements, and Troubleshooting Handbooks

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**Operational Warning**

Any attempt to systematically bypass remote connectivity firewalls, forge hardware identification strings, or share Multi-Factor Authentication session parameters will result in immediate termination of all directory credentials and referral to the Enterprise Risk Management board under zero-tolerance network manipulation provisions.

## 1.0 CORPORATE CONTEXT, SCOPE, AND PURPOSE

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### 1.1 Enterprise Security Imperative

AeroNexus Technologies operates a zero-trust network infrastructure across multiple public cloud availability zones, physical lab installations, and production clusters. Securing communication channels between remote workspaces and corporate core intellectual assets is a critical engineering requirement. The purpose of this guide is to delineate the strict technical configurations, cryptographic expectations, identity checks, and troubleshooting procedures required to govern all remote entry vectors.

### 1.2 Boundary of Scope

This technical guide applies directly to all connections passing through the ANX-SecureConnect platform gateways. This covers all regional developer environments, continuous integration nodes, internal accounting tools, and source repository systems. All regular corporate full-time employees (FTEs), external design agency integrations, third-party support vendors, and temporary technical operators must maintain full compliance with these standards.

### 1.3 Core Network Cryptography

The ANX-SecureConnect network framework exclusively utilizes automated AES-256-GCM data encryption protocols with ChaCha20-Poly1305 fallback options for approved lower-powered mobile endpoints. Every remote connection session enforces an automated ephemeral Diffie-Hellman key rotation mechanism every 60 minutes to eliminate replay vector exposure risks.

#### Regulatory Framework Alignment

The criteria detailed inside this standard manual are structured to fully guarantee compliance with ISO/IEC 27001:2022 Control A.8.20 (Network Security) and SOC 2 Type II Trust Services Criteria for security, confidentiality, and processing integrity.

## 2.0 ELIGIBILITY CRITERIA & DYNAMIC REQUEST LIFECYCLE

### 2.1 Base Access Entitlement

Access tokens for remote VPN connectivity are granted on a business-need baseline. Personnel assigned to designated office-bound operational profiles must file an explicit project justification case to receive a remote connection pass. Technical personnel designated as hybrid or full-remote inside the corporate directory are granted a baseline access provisioning tier automatically at onboarding.

### 2.2 Step-by-Step Remote Entitlement Workflow

To request access to specialized corporate network segments (such as the `*.prod-cluster.internal` target domain), personnel must adhere strictly to the following process mapping:

1. Navigate to the enterprise service system portal and select ticket workflow ANX - IAM - VPN - REQ.
2. Define the exact target subnets required, specifying requested duration windows (e.g., standard rolling 180-day assignment).
3. Input a verified business justification, citing active Jira project tokens or infrastructure architecture change models.
4. Submit the ticket to trigger automated endpoint posture discovery scans via the endpoint check agent.

| Access Tier                 | Target Network Boundary           | Primary Approver        | Secondary Clearance                     |
|-----------------------------|-----------------------------------|-------------------------|-----------------------------------------|
| Tier 1: Basic Corporate     | Intranet, Email, Jira, Confluence | Automated Posture Check | System Validation Complete              |
| Tier 2: Engineering Staging | Non-Production Labs, Git Repos    | Direct Line Manager     | Security Operations Clearance           |
| Tier 3: Production Core     | Live Databases, Transaction Tiers | Engineering Director    | VP of Security Operations (Veto Option) |

## 3.0 IDENTITY ASSURANCE & MULTI-FACTOR AUTHENTICATION STANDARDS

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### 3.1 Universal Multi-Factor Authentication (MFA) Mandate

Access through the ANX-SecureConnect client interface requires active verification via our centralized identity engine (ANX-Identity). Simple password validation strings are never accepted as a sufficient remote connection challenge vector under any operational scenario.

### 3.2 Approved Authentication Verification Pathways

The enterprise identity boundary strictly supports three isolated cryptographic validation pathways:

- **FIDO2 WebAuthn Hardware Keys:** Corporate-issued YubiKey 5 Series tokens mapped to the user profile footprint. This is the only approved authentication method for Tier 3 production access loops.
- **ANX-Identity Mobile Authenticator App:** Time-based One-Time Password (TOTP) structures or secure biometric push challenge approvals with mandatory number-matching validation checks.
- **Biometric Platform Verification:** Integrated Apple TouchID or Windows Hello Enterprise keys cryptographically linked to the specific TPM 2.0 component inside corporate laptops.

#### BANNED AUTHENTICATION DELIVERY MECHANISMS

The application of short message service (SMS-based text tokens) or standard voice call challenge confirmations is completely banned for remote identity validation. Any legacy profiles discovered using SMS-based profiles will be locked automatically by the identity sweep script at midnight.

#### SUPPORT SCENARIO: FATIGUE PUSH MITIGATION RESPONSE

An engineer experiences an MFA push flooding attack, receiving 14 sequential push verification challenges on their phone device at 2:00 AM. The engineer ignores the pushes and reports the anomaly via the corporate phone emergency desk. Under section 3.2, the identity system flags the anomaly, instantly quarantines the primary directory account, and invalidates all active VPN session tokens globally.

## 4.0 APPROVED PLATFORM SIZING & DEVICE SPECIFICATIONS

### 4.1 Standard Managed Host Systems

The corporate client application (ANX-SecureConnect Client v4.6+) runs exclusively on systems maintaining verified hardware abstraction layer parameters. Operating systems must be managed active instances integrated with the corporate Mobile Device Management (MDM) endpoint controller platform.

| Operating System Platform    | Minimum Build Level         | Patch Level Requirement                  | Required Local Security Agent |
|------------------------------|-----------------------------|------------------------------------------|-------------------------------|
| Microsoft Windows Enterprise | Windows 11 (Build 23H2+)    | Latest Cumulative Update (≤ 15 days old) | ANX-EndpointShield v6.2       |
| Apple macOS Enterprise       | macOS 14.4 (Sonoma) +       | Current Stable Sub-version Release       | ANX-EndpointShield v6.2       |
| Enterprise Linux Pods        | Ubuntu LTS 24.04 / RHEL 9.3 | Current Repository Security Rollups      | ANX-LinuxAudit Daemon         |

### 4.2 Client Automation Configurations

The client installation package includes automated background posture daemons that execute a hardware check sweep prior to initializing the encapsulation tunnel loop. The host hardware framework must possess an active, functional Trusted Platform Module (TPM v2.0) chip module or an Apple Secure Enclave processor core to hold encryption token material securely inside local memory storage partitions.

#### Mobile Operating System Access Policy

Mobile operating systems (Apple iOS and Google Android) are restricted exclusively to basic communications applications (Slack and Outlook). Direct command-line shell access, secure shell tunneling, or repository pulling through the VPN application on mobile form factors is strictly blocked by default.

## 5.0 CORPORATE BRING-YOUR-OWN-DEVICE (BYOD) BOUNDARIES

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### 5.1 The BYOD Remote Provisioning Perimeter

To support agile workspace allocation, AeroNexus Technologies permits partial BYOD network access loops for specific corporate operations. However, personal unmanaged machines are structurally barred from establishing direct routing connections into the enterprise core network. Instead, BYOD endpoints connect exclusively to an isolated Virtual Desktop Infrastructure container environment (ANX-CloudWorkplace).

### 5.2 Strict Container Boundaries and Isolation Rules

When connecting to corporate systems from a personal BYOD computer terminal, the following operational boundaries are enforced via automated virtualization client rules:

- **Local Storage Isolation:** Downloading files from the corporate remote cloud container onto the local personal drive space is completely blocked by system policy.
- **Clipboard Control Restrictions:** Bi-directional text copy-pasting functionalities between the corporate workspace container and the host personal operating system are restricted to 500 characters max, and binary code copy operations are completely blocked.
- **Peripheral Device Blocking:** Local personal peripheral storage hardware, home printing infrastructure elements, and secondary web camera feeds are barred from passing telemetry pipelines into the secure container space.

#### SUPPORT SCENARIO: BYOD SESSION TERMINATION ACTION

A software testing vendor logs into their virtual space from a personal desktop. At the same time, a personal background screen recording tool runs on the host personal desktop space. The virtual workspace client detects the active system-level virtualization injection hook, drops the active connection instantly, and logs incident flag `ANX-BYOD-SCREENCAP` with the security ops team.

## 6.0 ENDPOINT TECHNICAL SECURITY REQUIREMENTS

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### 6.1 Pre-Connection Host Health Verification

Prior to authenticating an incoming session request, the ANX-SecureConnect system executes an automated host health posture check script. If the remote endpoint lacks any of the mandatory security baseline states, the connection attempt is denied instantly, and the system forwards a remediation profile code snippet to the client log interface.

### 6.2 Mandated Endpoint Posture Baselines

The remote device must pass four explicit verification criteria loops to satisfy security constraints:

1. **Full Disk Encryption (FDE):** Windows BitLocker or macOS FileVault must be active across all primary and secondary storage volumes, enforcing minimum AES-256 structural strength.
2. **EDR Health Verification:** The ANX-EndpointShield agent must be online, communicating with corporate security hubs, and running update definitions less than 24 hours old.
3. **Local Firewall Enforcement:** The native system firewall profile must be actively configured to block all incoming connection packages from public network paths.
4. **Process Isolation Compliance:** No active network traffic capture packages (such as Wireshark or tcpdump) or uncertified execution binaries can be active inside the process tracking list during connection initialization.

#### STRICT GEOFENCING RESTRICTIONS

The corporate remote access framework drops all incoming connection packages originating from geopolitical zones defined as high-risk by the Enterprise Risk Management board. Personnel traveling cross-border must register a pre-authorized flight profile ticket (`ANX-TRAVEL-SEC`) to bypass automated regional blocks.



## 7.0 STEP-BY-STEP INCIDENT RESOLUTION & COMMON ERROR CODE MATRIX

### 7.1 Common Client Error Definitions

| System Error Code | Root Cause Interpretation                            | Standard Resolution Protocol Steps                                                                                       |
|-------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ERR-ANX-4011      | Posture Assessment Failure: Disk Encryption Inactive | Open system settings panel, re-verify BitLocker/FileVault initialization progress, restart endpoint system.              |
| ERR-ANX-4089      | Cryptographic Clock De-synchronization Anomaly       | Navigate to system time settings, toggle dynamic network time synchronization option on, point clock to target NTP pool. |
| ERR-ANX-5003      | Gateway Peer Handshake Failure (UDP Timeout Block)   | Verify local router config allows UDP port 4500 traffic; switch interface to TCP Over-encapsulation Fallback Mode.       |
| ERR-ANX-6012      | TPM Validation Challenge Invalidation                | Flush systemic identity credentials, execute hardware chip re-initialization sequence inside local system firmware.      |

### 7.2 Tiered Support Escalation Matrix

If standard remediation pathways fail to resolve client connection blocks, personnel must escalate issues strictly matching the following matrix parameters:

- **Tier 1 Support (IT Desk Helpdesk):** Initial ticket logging via portal or extension 4400. Handles password resets, software reinstalls, and clear-cut error codes. Target response:  $\leq 2$  Hours.
- **Tier 2 Support (Endpoint Security Ops):** Triggered when ERR-ANX-6012 or structural posture verification bugs occur. Requires device review via MDM dashboard tools. Target resolution:  $\leq 4$  Hours.
- **Tier 3 Support (Core Network Infrastructure Team):** Triggered for broad regional routing gateway failures, cluster certificate errors, or active corporate firewall blocks. Target resolution: Immediate Action.

### 7.3 Access Revocation Context

Remote network credentials face automated systemic revocation if an account remains entirely inactive for 45 consecutive calendar days. Furthermore, immediate privilege revocation occurs upon formal HR separation notification or security breach identification alerts.